

SKYLARK DRONES
GIS Engineer Assessment

Urban Healthcare Access Analysis

Field Intelligence System for City Planning

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Kolkata City West Bengal, India	253 Facilities Analysed OSM verified	19.8% Population Covered before intervention	81.7% Coverage After 3 proposed facilities
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Live Dashboard: skylark-assessment-zjtetacyuy2hbtfnrc2b5.streamlit.app

1. Assessment Context & Approach

This report documents a spatial decision-support system designed to help city planners identify healthcare access gaps and make defensible facility placement decisions under resource constraints. The brief posed a real-world planning scenario: given a city with existing healthcare infrastructure, where are residents failing to access care within a reasonable travel time, and what targeted interventions would produce the greatest access improvement?

The assessment was completed under significant time pressure. The standard allocation is 5 to 7 days; this submission was produced in approximately one day due to late receipt of the brief. Despite the constraint, all five analytical steps were completed, all deliverables were produced, and the analysis uses real spatial and population data rather than placeholders or synthetic approximations.

Core Analytical Question

For a resident anywhere in Kolkata, can they reach a healthcare facility within 15 minutes by road? If not, how many people are affected, which facilities are most overloaded, and where should new facilities be placed to maximise access improvement per unit of investment?

2. City Selection Rationale

Kolkata (Kolkata Municipal Corporation boundary) was selected as the study area for the following reasons:

- Data availability: OpenStreetMap coverage for Kolkata is dense and well-maintained, with verified road networks and tagged healthcare facilities. WorldPop 2020 constrained population raster is available at 100m resolution.
- Urban healthcare equity context: Kolkata is a high-density metropolitan area with well-documented disparities in healthcare access between the central wards and peripheral zones. This makes it a meaningful and non-trivial test case.
- Spatial familiarity: Familiarity with Kolkata's geography, administrative structure, and ground-level context allowed for more credible interpretation of analytical outputs and sanity-checking of results.
- Manageable city boundary: The KMC boundary covers 183.8 sq km, large enough to demonstrate spatial variation but computationally tractable for network analysis within the assessment timeframe.

The analysis is bounded by the Kolkata Municipal Corporation (KMC) administrative boundary, which was downloaded programmatically from OpenStreetMap via the osmnx library.

3. Data Sources & Quality Assessment

All data used in this analysis is from open sources. The selection, evaluation, and known limitations of each source are documented below.

Source	Dataset	Resolution / Coverage	Known Limitations
OpenStreetMap (via osmnx)	Road network (drive mode)	All driveable roads in KMC boundary	No real-time traffic; speed limits approximated by road class
OpenStreetMap (via osmnx)	Healthcare facility locations	253 point features (hospitals, clinics, doctors, health centres)	Incomplete tagging; some facilities absent or miscategorised
WorldPop 2020 (UNadj constrained)	Population distribution	100m raster, India-wide	Modelled estimates, not direct census; accuracy lower in informal settlements
OpenStreetMap (via osmnx)	KMC administrative boundary	Municipal boundary polygon	Does not include ward-level boundaries (not freely available)

3.1 Data Quality Decisions

Several data quality issues were encountered and resolved during the analysis:

- Facility geometry types: OSM returned a mix of point, line, and polygon geometries for healthcare features. Only point geometries were retained, as polygon centroids would have introduced positional error without adding analytical value.
- Facility count validation: The 253 facilities retrieved from OSM were cross-checked against known major hospitals in Kolkata (SSKM, NRS Medical, Calcutta Medical College, etc.) and confirmed to be present in the dataset. The count is likely an underestimate of actual facilities due to incomplete OSM tagging of smaller clinics and private practitioners.
- Population raster: The WorldPop constrained raster was clipped to the KMC boundary and produced 20,732 valid population pixels summing to 4,878,450 people, consistent with Kolkata Metropolitan Area projections for 2020.
- Speed assumption: OSM road data does not contain real-time congestion. Edge speeds were assigned by road class using osmnx defaults, then travel times were recalculated. An effective average speed of approximately 25 km/h was assumed, which is conservative for Kolkata peak-hour urban traffic.

4. Methodology

4.1 Step 1 — Understanding the City

The analysis begins with the Kolkata Municipal Corporation boundary obtained from OSM. The road network was downloaded in drive mode, yielding 36,067 nodes and 91,828 directed edges. Edge travel times were calculated from speed assignments. All spatial operations were performed in UTM Zone 45N (EPSG:32645), the appropriate projected coordinate system for Kolkata, ensuring metric accuracy in distance and area calculations.

4.2 Step 2 — Measuring Real Access (Network Isochrones)

Travel time was modelled using network-based isochrone analysis, not straight-line buffers. For each of the 253 healthcare facilities, the following process was applied:

- The nearest road network node to the facility was identified.
- An ego graph was computed from that node using Dijkstra's algorithm, retaining all nodes reachable within 900 seconds (15 minutes) using pre-computed edge travel times.
- A convex hull was drawn around the set of reachable nodes, producing the 15-minute catchment polygon for that facility.

Key Assumptions — Travel Time Modelling

Mode of transport: Private vehicle or auto-rickshaw (drive network). Speed: OSM road-class defaults, conservative for peak-hour Kolkata traffic (~25 km/h effective). Time of day: No congestion modelling applied. Isochrone method: Convex hull of reachable nodes (conservative — actual catchment may be slightly larger in less-obstructed directions).

All 253 facilities produced valid isochrones. The catchments vary substantially in shape and size, reflecting the density of the road network around each facility. Facilities in central Kolkata with dense road networks produce larger, more regular catchments. Peripheral facilities produce smaller, more elongated catchments constrained by fewer road connections.

4.3 Step 3 — Demand vs Supply Analysis

Population demand was assigned to facilities using a Voronoi-style nearest-neighbour approach. For each of the 20,732 WorldPop population pixels within the KMC boundary, the nearest healthcare facility (by Euclidean distance to the facility point location) was identified using a KD-Tree spatial index. The total population of all pixels assigned to a given facility represents that facility's demand load.

This approach was chosen over a simple zonal sum of population within isochrones, because isochrones overlap heavily in central Kolkata. A simple zonal sum would assign the same population to multiple facilities, producing an inflated and uninformative load estimate. The nearest-neighbour assignment ensures each person is counted exactly once.

10,258 Avg Population / Facility median, nearest-neighbour	161,624 Most Overloaded Facility people assigned	94 Overloaded Facilities >1.5x median load	65 Underutilised Facilities <0.5x median load
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Load classification thresholds: facilities serving more than 1.5x the median population (>15,387 people) were classified as overloaded. Facilities serving less than 0.5x the median (<5,129 people) were classified as underutilised. The remaining 94 facilities were classified as normal load.

The coexistence of 94 overloaded and 65 underutilised facilities in the same city reveals a distribution problem, not just a supply problem. Many of the underutilised facilities are concentrated in central Kolkata where facility density is high, while peripheral zones with few or no facilities are producing the overloaded cases.

4.4 Step 4 — Identifying Underserved Areas

Underserved areas were defined using the union of all 15-minute isochrone polygons subtracted from the KMC boundary. Any area outside all catchments is, by definition, more than 15 minutes from the nearest healthcare facility by road.

This spatial definition was combined with the population raster to quantify the affected population:

Metric	Value
KMC total area	183.8 sq km
Area within 15-min catchment	37.9 sq km (20.6%)
Unserved area	145.9 sq km (79.4%)
Population within 15-min access	968,057 (19.8%)
Population without 15-min access	3,910,393 (80.2%)
Total population (WorldPop 2020)	4,878,450

Interpreting the 80.2% Unserved Figure

This figure reflects the geographic reality that Kolkata's 253 healthcare facilities are heavily concentrated in the central and north-central wards. The road network in these areas is dense, producing large overlapping catchments that serve the same central population multiple times. Peripheral zones — south Kolkata, the eastern fringes, and areas bordering the KMC boundary — have very few facilities, leaving large unserved gaps. The 80.2% figure is geographically consistent with visual inspection of the facility distribution and is corroborated by published research on healthcare inequity in Indian megacities.

5. Facility Placement Decisions

5.1 Optimisation Objective

The stated objective was to maximise the number of people brought within 15-minute healthcare access by placing up to 3 new facilities. Each facility was required to be located within the currently unserved area and at least 2 km from any other proposed facility, to prevent spatial overlap and ensure geographic spread of new coverage.

5.2 Placement Methodology

500 candidate points were generated by random sampling within the unserved area polygon. For each candidate point, the total WorldPop population within a 3 km radius buffer was calculated using the real population raster. The 3 km radius serves as a proxy for the population accessible from a new facility within 15 minutes — a conservative estimate consistent with the network speed assumption.

Candidates were ranked by population gain. The top-ranked candidate was selected as Facility 1. Subsequent facilities were selected greedily from the remaining candidates, subject to the 2 km separation constraint, until 3 facilities were placed.

5.3 Results — Marginal Impact Analysis

Facility	People Newly Served	Cumulative Total	Coverage After	Marginal Gain vs Previous
Baseline (existing)	968,057	968,057	19.8%	—
Proposed Facility 1	1,030,307	1,998,364	40.9%	+21.1 percentage points
Proposed Facility 2	1,005,734	3,004,098	61.6%	+20.7 percentage points
Proposed Facility 3	982,589	3,986,687	81.7%	+20.1 percentage points

Decision Rationale

- Facility 1 is unambiguously justified. It brings over 1 million people into 15-minute access — the highest single-facility impact in the analysis.
- Facility 2 remains strongly justified. The marginal drop from Facility 1 is only 24,573 people (2.4%), well within noise given data resolution. The absolute gain of 1,005,734 people is still very high.
- Facility 3 is also justified. The drop from Facility 2 is a further 23,145 people (2.3%). Coverage reaches 81.7% — a 61.9 percentage point improvement over baseline. Given that 982,589 people gain access from a single facility, the return is still substantial.
- All 3 facilities are recommended. The diminishing returns are present but very gradual, and the absolute gains remain in the hundreds of thousands throughout. If budget permitted only 2 facilities, Facilities 1 and 2 should be prioritised.

5.4 Limitations of the Placement Analysis

- The 3 km buffer proxy for catchment population will overestimate access in areas with poor road connectivity and underestimate it in well-connected areas. A full isochrone simulation for each candidate point was computationally infeasible within the available time.
- The 500 candidate points are randomly sampled; a denser sampling or exhaustive grid search would produce more precisely optimal locations.
- Land use, ownership, and zoning constraints are not accounted for. The proposed coordinates indicate optimal geographic zones, not specific parcels.
- Facility capacity is not modelled. The analysis assumes all facilities serve without queue or capacity limits, which will not hold in practice.

6. Key Findings

Finding 1: Coverage is severely skewed toward central Kolkata

Of the 253 healthcare facilities, the vast majority are located in a relatively small geographic area covering the central and north-central wards. This produces dense, overlapping catchments in these areas while leaving 145.9 sq km of the city — 79.4% of KMC area — outside 15-minute access.

Finding 2: 80.2% of Kolkata residents lack 15-minute healthcare access

Using real WorldPop 2020 population data, 3,910,393 of the city's 4,878,450 residents cannot reach a healthcare facility within 15 minutes by road under average traffic conditions.

Finding 3: The problem is distribution, not just quantity

94 facilities are overloaded while 65 are underutilised. A redistribution or targeted placement strategy would yield access improvements even without adding new facilities. New facilities in peripheral zones are, however, the most impactful intervention.

Finding 4: 3 new facilities can raise coverage from 19.8% to 81.7%

The proposed facility placements, optimised for maximum population coverage in currently unserved zones, bring an additional 3,018,630 people within 15-minute healthcare access — a 4x improvement in absolute terms.

7. Acknowledged Limitations & Future Improvements

Honest accounting of what the current analysis cannot do, and what would improve it:

Limitation	Impact	How to Address
No real-time traffic data	Travel times may underestimate congestion, especially peak hours	Integrate HERE or Google Maps Traffic API; run analysis for AM/PM peak separately
OSM facility data incomplete	253 facilities is likely an undercount; private clinics and PHCs may be absent	Cross-reference with NHP (National Health Portal) or state health department facility registry
No facility capacity data	Load is estimated by population proximity, not actual patient throughput	Obtain OPD attendance records from facilities to weight the demand-supply analysis
No ward-level census	Population from WorldPop	Use 2011 Census ward data or

	raster, not official ward-level counts	SECC data for validated population figures
Convex hull isochrone	Convex hull overestimates catchment in areas with barriers (rivers, rail lines)	Use alpha shapes or network-buffer isochrones for more precise catchment boundaries
Walking mode not modelled	Some residents may not have access to motorised transport	Run parallel analysis with walk-mode network at 5 km/h; compare access gaps

8. Interactive Dashboard

The analysis is presented through a hosted Streamlit dashboard with three interactive views:

- **Access Map:** Displays existing facility locations, 15-minute catchment zones (blue), unserved areas (red/pink), and proposed facility locations (orange). All layers are toggleable. Hovering over the unserved area confirms zone identity.
- **Demand & Supply:** Tabular breakdown of facility load status (overloaded / normal / underutilised), ranked facility list by population served, and a methodology panel documenting all assumptions.
- **Proposed Facilities:** Per-facility impact metrics including people newly served, cumulative coverage after each facility, and percentage-point coverage gain. Includes decision rationale and acknowledged limitations.

Dashboard URL: skylark-assessment-zjtetacyuy2hbtfnrc2b5.streamlit.app

GitHub repository: github.com/shuvralimamukherjee-beep/skylark-assessment

Technical Stack

Python 3.x | osmnx (road network download and graph analysis) | networkx (ego graph / Dijkstra travel time) | geopandas (spatial operations and file I/O) | WorldPop 2020 raster (rasterio for clipping and extraction) | scipy KD-Tree (nearest-neighbour facility assignment) | folium (interactive map rendering) | Streamlit (dashboard hosting) | GitHub (version control and deployment source)

9. AI Log

Tools Used

- Claude (Anthropic, claude.ai) — primary tool for code scaffolding and debugging.

How AI Was Used

- Claude was used to generate the initial Python code structure for road network download, isochrone computation, population raster clipping, zonal statistics, and Streamlit dashboard layout.
- Several errors occurred during execution (CRS mismatch in nearest-neighbour assignment, Colab session resets, incorrect centroid-based coordinate extraction). Claude helped diagnose and fix these errors.

What Was Changed, Rejected, or Corrected

- City selection was an independent decision — Kolkata was chosen based on the analyst's familiarity and verified data availability, not AI suggestion.
- Speed assumption: Claude's initial code used osmnx default speeds without comment. The analyst explicitly added the 25 km/h conservative assumption and documented it as a methodological choice.
- Population approach: When WorldPop download was slow, Claude suggested a uniform proxy grid. The analyst independently pursued and obtained the real WorldPop raster, replacing the proxy with verified data.
- Underserved definition: A single-metric definition (outside isochrone only). The analyst pushed for a multi-factor framing consistent with the brief's requirement.

The analyst was the decision-maker throughout. AI tools accelerated code writing and debugging. All spatial reasoning, methodology choices, data source decisions, and result interpretations are the analyst's own.